

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (MMD)(CBCS) Degree Examinations
 January 2023

22MMD113 - MECHANISM DESIGN

Duration: 3 Hours

Max. Marks:100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Missing data, if any may be suitably assumed.

Unit – I

Marks BT* CO* PO*

1. a) Draw the unscaled LPE and calculate DOF for the original and LPE linkage shown in fig 1(a).

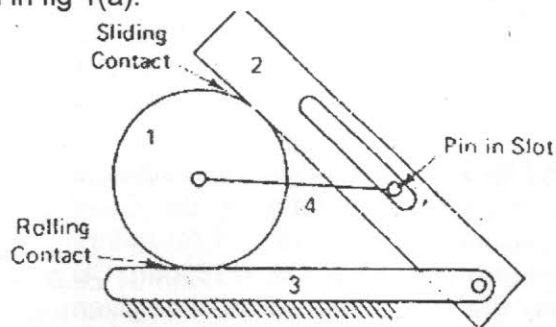


fig.1(a)

10 L*2 1 2

- b) Determine the transmission and deviation angle(s) for the mechanism shown in fig.1(b) by considering i) link 2 as input and ii) link 6 as input.

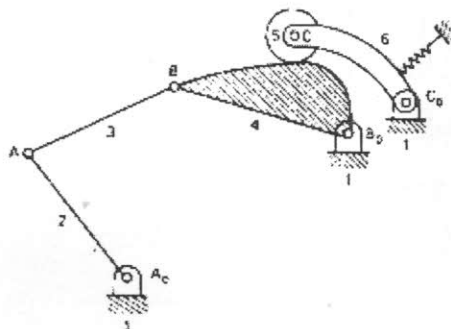


Fig: 1(b)

10 L2 1 2

2. a) Briefly explain the following terminologies with an example each:
- i. Transmission Angle
 - ii. Flexible link
 - iii. Form closed pair
 - iv. Inversion
 - v. Rigid body guidance

10 L1 1 1

- b) Fig. 2(b) shows the whit-worth quick return mechanism used in reciprocating machine tools. The various dimensions are: $OQ=100\text{mm}$, $OP=200\text{mm}$, $RQ=150\text{mm}$, $RS=500\text{mm}$. The crank OP makes an angle of 60° with the vertical. Determine the velocity of slider S when the crank rotates at a speed of 120rpm CW. Also find the angular velocity of link RS and the velocity of the sliding block T on the slotted lever QT .

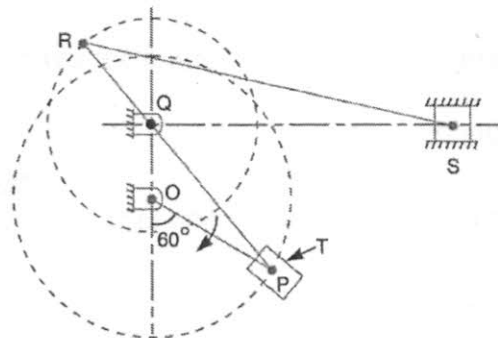


fig.2(b)

10 L3 2 2

3. In the toggle mechanism shown in Fig. 3, the slider D is constrained to move on a horizontal path. The crank OA is rotating in the CCW direction at a speed of 180 rpm , increasing at the rate of 50 rad/s^2 . The dimensions of the various links are as follows: $OA = 180\text{mm}$; $CB = 240\text{mm}$; $AB = 360\text{mm}$; and $BD = 540\text{mm}$. For the given configuration, find 1) Velocity of slider D and angular velocity of BD and, 2) Acceleration of slider D and angular acceleration of BD .

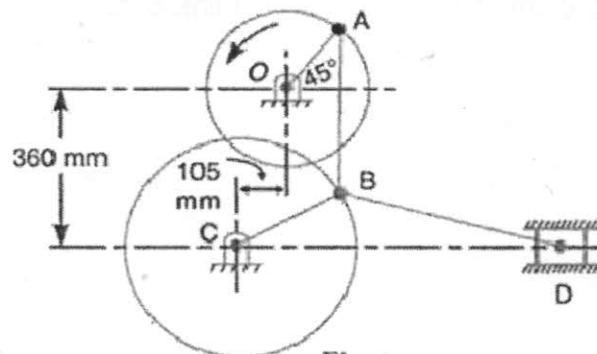


Fig.3

20 L3 2 2

Unit – II

4. a) Briefly explain the procedure for two position synthesis of a crank-rocker mechanism when the time ratio is given, with an example.
 b) A 4 bar linkage is required to generate a function $y=x^{1.6}$ for $1 \leq x \leq 3$. The crank rotates from an angle 60° to 120° whereas follower rotates from an angle of 60° to 150° . Find: $y_1, y_2, y_3, \phi_j, \psi_j$ ($j=2,3$) using Chebyshev's spacing.
5. a) Briefly explain the procedure for two position synthesis of a centered and offset slider crank mechanism with neat drawing.

10 L2 3 1

10 L3 3 2

10 L1 3 1

- b) In the fig.5(b), $AB = 1.25$ cm. Use A and B as circle points, and design a four-bar linkage to move its coupler through the three positions shown. Use Grashof's equation to identify the type of four bar linkage designed.

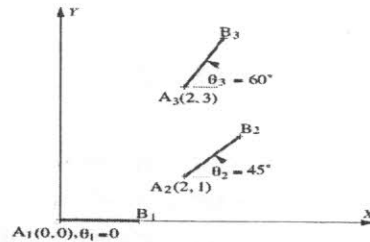


fig.5(b)

10 L3 4 2

6. Design a mechanism using Freudenstein's equation to generate function $y = \log_{10}(x)$ for $1 \leq x \leq 2$. The input and output link range is $\Delta\theta = 90^\circ$ and $\Delta\psi = 60^\circ$ both CCW respectively. Take $\phi_s = 60^\circ$ and $\psi_s = 60^\circ$ be the starting value for input and output links. Take 3 accuracy points and assume frame link length 5cm.

20 L4 4 2

Unit – III

7. a) The crank-pin circle radius of a horizontal engine is 300 mm. The mass of the reciprocating parts is 250 kg. When the crank has travelled 60° from IDC., the difference between the driving and the back pressures is 0.35 N/mm^2 . The connecting rod length between centres is 1.2 m and the cylinder bore is 0.5 m. If the engine runs at 250 rpm. and if the effect of piston rod diameter is neglected, calculate: 1) pressure on slide bars, 2) thrust in the connecting rod, 3) tangential force on the crank-pin, and 4) turning moment on the crank shaft.
- b) The 4 bar mechanism as shown in fig.7(b). has one external force $P = 200 \text{ N}$ and one inertia force $S = 150 \text{ N}$ acting on it. The system is in dynamic equilibrium as a result of torque T_2 applied to link 2. Find T_2 and pin forces.

10 L3 5 2

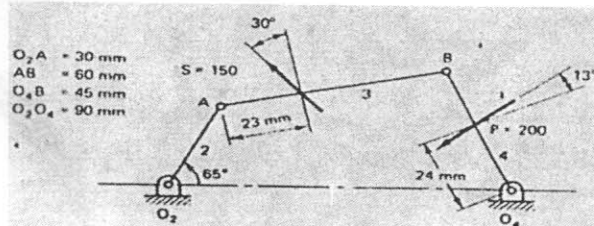


fig.7(b)

10 L4 5 2

8. a) The crank and connecting rod of a steam engine are 0.3 m and 1.5 m in length. The crank rotates at 180 rpm clockwise. Determine the velocity and acceleration of the piston when the crank is at 40 degrees from the inner dead centre position. Also determine the position of the crank for zero acceleration of the piston.
- b) A slider crank mechanism with the following dimensions is acted upon by a force $F = 2 \text{ kN}$ at B as shown in fig 8(b): $OA = 100 \text{ mm}$, $AB = 450 \text{ mm}$. Determine the input torque T on the link OA for static equilibrium of the mechanism.

10 L3 5 2

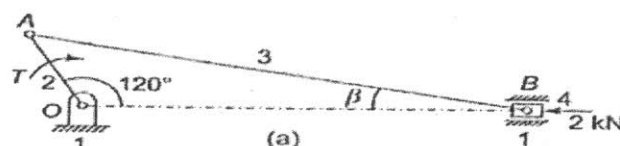


Fig 8(b)

10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
